

AI Vendor Risk Investigation Agent

Autonomous Multi-Agent Enterprise Risk Intelligence Platform

1. Executive Summary

AI Vendor Risk Investigation Agent is an autonomous multi-agent AI platform designed to modernize enterprise vendor on-boarding and compliance verification.

The platform uses multiple specialized AI agents that collaborate together to:

- Validate vendors
- Analyze compliance documents
- Detect duplicate or blacklisted vendors
- Verify GST and website information
- Assess vendor credibility
- Generate explainable AI risk scores
- Produce automated investigation reports

The solution combines:

- Microsoft Copilot Studio
- AI Builder
- Power Automate
- SharePoint
- Computer Use
- Multi-Agent Orchestration

to create an enterprise-grade autonomous investigation platform.

2. Problem Statement

Enterprises globally face significant operational and financial risks during vendor on-boarding.

Traditional vendor verification processes are highly manual and involve:

- Validating GST registrations
- Reviewing on-boarding documents
- Checking blacklisted vendors
- Verifying websites and digital presence
- Assessing compliance requirements
- Creating investigation reports manually

This process creates several challenges:

Challenge	Impact
Manual Verification	High operational effort
Vendor Fraud	Financial and compliance risk
Duplicate Vendors	Procurement inefficiencies
Human Errors	Incorrect approvals
Slow Investigations	Delayed onboarding
Lack of Auditability	Governance gaps

Organizations can lose millions annually due to:

- Fake suppliers
- Fraudulent on-boarding
- Compliance violations
- Poor verification processes

There is a need for an intelligent, scalable, and autonomous vendor risk investigation platform.

3. Proposed Solution

AI Vendor Risk Investigation Agent is a multi-agent AI platform that autonomously performs vendor investigations.

The solution supports two operating modes:

Mode 1 – Event-Driven Automation

The workflow automatically triggers whenever a vendor onboarding email is received.

Mode 2 – Interactive Copilot Experience

Users can directly interact with the AI system through Copilot chat.

The platform orchestrates multiple AI agents that independently investigate different dimensions of vendor risk.

The final outcome includes:

- Vendor verification status
- AI-generated risk score
- Explainable reasoning
- Investigation summary
- Compliance recommendations

4. Key Features

1. Autonomous Multi-Agent Investigation

Multiple AI agents collaborate together to investigate vendor credibility.

2. Computer Use Automation

AI agents autonomously interact with external websites such as GST portals.

3. AI-Powered Document Intelligence

Extracts and validates information from onboarding documents.

4. Blacklist & Duplicate Detection

Prevents duplicate and fraudulent vendor onboarding.

5. Explainable AI Risk Scoring

Provides transparent vendor risk analysis.

6. Event-Driven Workflow Automation

Automatically processes vendor onboarding requests.

7. Human-in-the-Loop Governance

Allows compliance officers to review AI-generated recommendations.

8. Enterprise Auditability

Centralized storage and traceable investigation history.

5. Innovation Highlights

1. Multi-Agent Architecture

Unlike traditional AI assistants that rely on a single centralized model, the AI Vendor Risk Investigation Agent uses a specialized **Multi-Agent Architecture**. The platform consists of multiple autonomous AI agents, where each agent is responsible for a dedicated investigation task such as blacklist detection, document intelligence, GST verification, website validation, and AI risk analysis.

These agents collaborate together through an orchestration layer to complete the end-to-end vendor investigation workflow. This modular architecture improves scalability, maintainability, reliability, and task specialization. Instead of behaving like a generic chatbot, the system operates as a coordinated team of AI investigators capable of handling complex enterprise verification processes.

2. Computer Use Integration

One of the most innovative capabilities of the platform is the integration of Computer Use in Microsoft Copilot Studio. Many enterprise verification tasks still require employees to manually interact with external government websites and portals that do not expose APIs.

The platform automates these browser-based verification activities using Computer Use. The AI agent can autonomously:

- * open a browser,
- * navigate to GST portals,
- * enter vendor GST numbers,
- * retrieve verification details,
- * and return structured results to the workflow.

This transforms previously manual operational tasks into autonomous AI-driven processes, significantly reducing manual effort, improving accuracy, and accelerating vendor onboarding operations.

3. Explainable AI

The platform incorporates **Explainable AI** capabilities to ensure transparency and enterprise trust. Instead of only generating a vendor risk score, the system also explains the reasoning behind the assessment.

The AI Risk Scoring Agent aggregates findings from all investigation agents and analyzes multiple risk indicators such as:

- * duplicate vendor detection,
- * blacklist presence,
- * GST mismatches,
- * invalid onboarding documents,
- * and suspicious digital presence.

The system then generates explainable outputs describing why a vendor has been classified as Low, Medium, or High Risk. This enables compliance teams and procurement officers to understand AI decisions clearly, improving governance, accountability, and responsible AI adoption.

4. Enterprise Governance Focus

The solution is specifically designed for real-world enterprise governance, procurement, and compliance scenarios. Unlike generic AI assistants, the platform focuses on operational challenges commonly faced by enterprises during vendor onboarding and verification.

The platform supports:

- * centralized audit trails,
- * human approval workflows,
- * role-based operations,
- * structured investigation tracking,
- * and compliance-oriented decision-making.

By integrating with enterprise systems such as Outlook, SharePoint, Power Automate, and Copilot Studio, the platform enables organizations to operationalize AI-driven governance directly within existing business processes.

5. Autonomous Investigation Workflow

The AI Vendor Risk Investigation Agent operates as an ****Autonomous Investigation Platform**** capable of independently performing vendor verification activities across multiple systems and processes.

Once a vendor onboarding request is received, the system automatically:

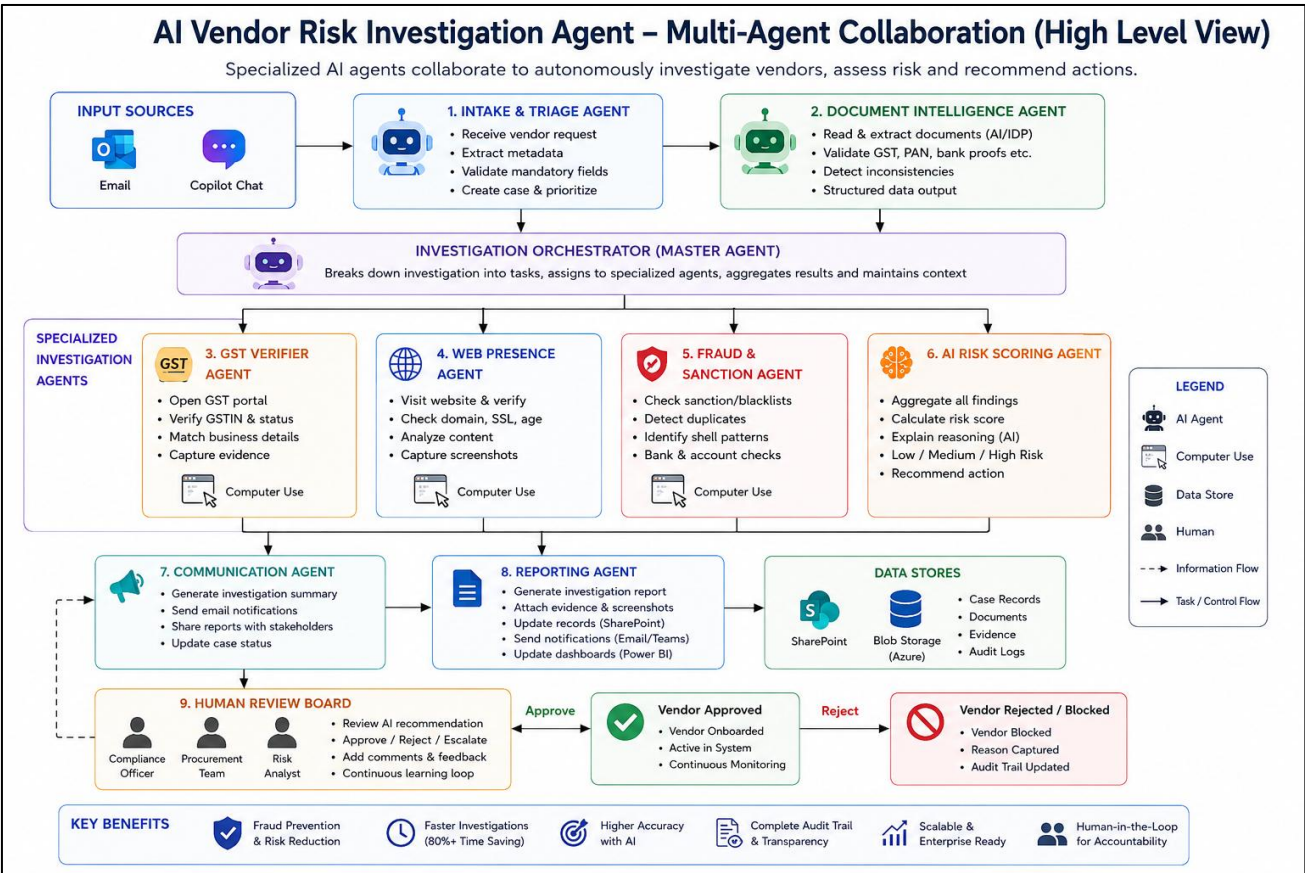
- * extracts vendor information,
- * stores onboarding documents,
- * initiates multi-agent investigations,
- * verifies GST registrations,
- * validates websites,
- * analyzes compliance documents,
- * calculates vendor risk,
- * and generates investigation reports.

This autonomous workflow reduces dependency on manual operations while enabling faster, more scalable, and more consistent enterprise investigations. The platform demonstrates how AI-driven orchestration can modernize enterprise operational intelligence and compliance verification processes.

6. System Architecture

Core Components

Component	Responsibility
Outlook Trigger	Starts onboarding workflow
Power Automate	Workflow orchestration
SharePoint	Centralized document storage
Copilot Studio	AI orchestration and agents
AI Builder	Document intelligence
Computer Use	Browser automation
Risk Engine	AI scoring and recommendations
Notification Layer	Final reporting



7. Multi-Agent Collaboration Architecture

The platform uses specialized AI agents coordinated by an orchestration layer.

Agents Used

1. Intake Agent

- Receives on-boarding requests
- Extracts vendor metadata
- Initializes investigation workflow

2. Blacklist & Duplicate Detection Agent

- Checks existing vendor records
- Detects duplicate vendors
- Detects blacklisted entities

3. Document Intelligence Agent

- Reads uploaded onboarding documents
- Extracts structured information
- Validates consistency

4. GST Verification Agent

- Uses Computer Use
- Navigates GST portal
- Verifies GST registration
- Extracts verification details

5. Web Presence Agent

- Verifies vendor website
- Validates digital presence
- Checks accessibility and credibility

6. AI Risk Scoring Agent

- Aggregates findings from all agents
- Generates vendor risk score
- Produces explainable reasoning

7. Communicator Agent

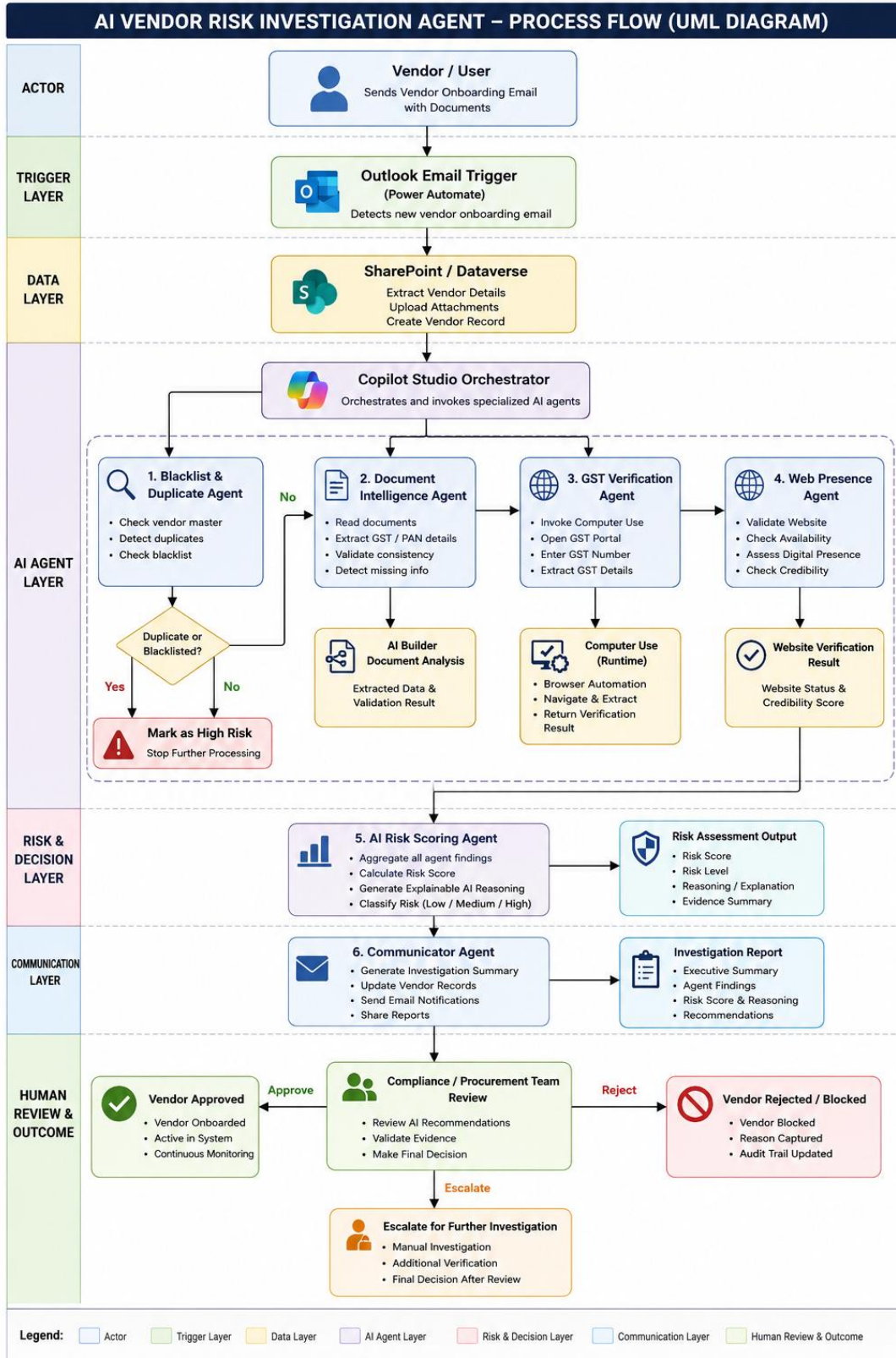
- Generates investigation summary
- Sends email notifications
- Shares final verification results



+ Add an agent

Name	Relationship	Trigger
 AI Risk Scoring Agent	 Child	 By agent
 Backlists And Duplicate Detector Agent	 Child	 By agent
 Communication Agents	 Child	 By agent
 Document Intelligence Agent	 Child	 By agent
 GST Verifier Agent	 Child	 By agent
 Web Presence Agent	 Child	 By agent

9. End-to-End Workflow



Step 1 – Vendor Email Received

A vendor onboarding request is received through email.

Step 2 – Trigger Execution

Power Automate automatically triggers the workflow.

Step 3 – Document Upload

Attachments are uploaded to SharePoint.

Step 4 – Multi-Agent Investigation

Specialized AI agents independently perform investigations.

Step 5 – Computer Use Verification

AI agents autonomously verify GST details on government portals.

Step 6 – AI Risk Assessment

The AI Risk Scoring Agent calculates vendor risk.

Step 7 – Investigation Report

Final investigation summary is generated.

Step 8 – Human Review

Compliance teams validate the final recommendation.

10. AI Agent Responsibilities

AI Agent	Key Responsibility
Intake Agent	Process onboarding request
Duplicate Detection Agent	Identify duplicate vendors
Blacklist Agent	Detect blocked vendors
Document Intelligence Agent	Validate onboarding documents
GST Verification Agent	Verify GST using Computer Use
Web Presence Agent	Validate website credibility
Risk Scoring Agent	Generate AI risk assessment

AI Agent	Key Responsibility
Communicator Agent	Send final report

11. Tools used

AI Vendor Risk Investigation Agent 🟢			
Overview Knowledge Tools Agents Topics Activity Evaluation Analytics +1			
+ Add a tool All Connector (2) Flow (2)			
Name	Type	Available to	Trigger
 Computer use - GST Verifications	 Computer use	 GST Verifier Agent	 By agent
 Computer use - Web Presence Check	 Computer use	 Web Presence Agent	 By agent
 Final Result Communicator	 Flow	 Communication Agents	 By agent
 Get Items - Vendors	 Connector	 Backlists And Duplicate Detector A...	 By agent
 Update item - Vendor Mater - Document analy...	 Connector	 Document Intelligence Agent	 By agent
 Vendor Document Analysis	 Flow	 Document Intelligence Agent	 By agent

10. Computer Use Integration

One of the key innovations of the platform is the use of Computer Use in Microsoft Copilot Studio.

Traditionally, employees manually:

- Open GST portals
- Enter GST numbers
- Verify registration details
- Copy investigation results

Using Computer Use, the AI agent autonomously:

- Opens the browser
- Navigates to the GST website
- Enters vendor GST number
- Retrieves official details

- Returns structured results

Benefits

Benefit	Impact
Reduced Manual Effort	Faster verification
Improved Accuracy	Reduced human error
Scalable Operations	Higher onboarding capacity
Autonomous Verification	Reduced dependency on manual teams

Details

What it is, how it operates, and how the orchestrator identifies it. [Learn more](#)

Name *

Computer use - GST Verifications

32/64

Description * ⓘ

GST Details Verification by Vendor Name

39/1024

Model

This model will be used for your computer-use tool. Experimental or preview models are subject to [preview terms](#). [Learn more](#)

Claude Sonnet 4.5

Instructions

Test

Instructions builder

Instructions *

To verify and validate an Indian GST Number (GSTIN) from the official Indian Government GST portal and extract the registered business details.

- i. Open the GST verification website URL - <https://clearax.in/gst-number-search/> in a web browser
- ii. Enter the GST Number (GSTIN) - GST Input for Vendor in the search/verification field
- iii. Obtain the following key details from the webpage:
 - GST Number: The GST Identification Number (GSTIN) provided for verification.
 - Legal Name of Business: The officially registered legal entity name associated with the GSTIN.
 - Trade Name: The business or trade name registered under the GSTIN, if available.
 - GST Status: The current status of the GST registration (e.g., Active, Cancelled, Suspended).
 - Registration Date: The date on which the GST registration became effective.
 - Constitution of Business: The type of business entity (e.g., Proprietorship, Partnership, Private Limited Company).
 - Taxpayer Type: Classification of taxpayer under GST (e.g., Regular, Composition, SEZ, Casual Taxpayer).

2204/8000

Inputs

What the tool accepts in order to run. Inputs will be filled in the order shown.

Input name	Fill using	Value
GST Number gstNumber	Dynamically fill with AI	Customize

Machines

Where should your automation run? Each machine option has different capabilities and requirements.

Machine *

Hosted browser

Hosted browser works with public websites only.
For desktop apps or private/internal sites, you'll need to create a new machine or select an existing one.

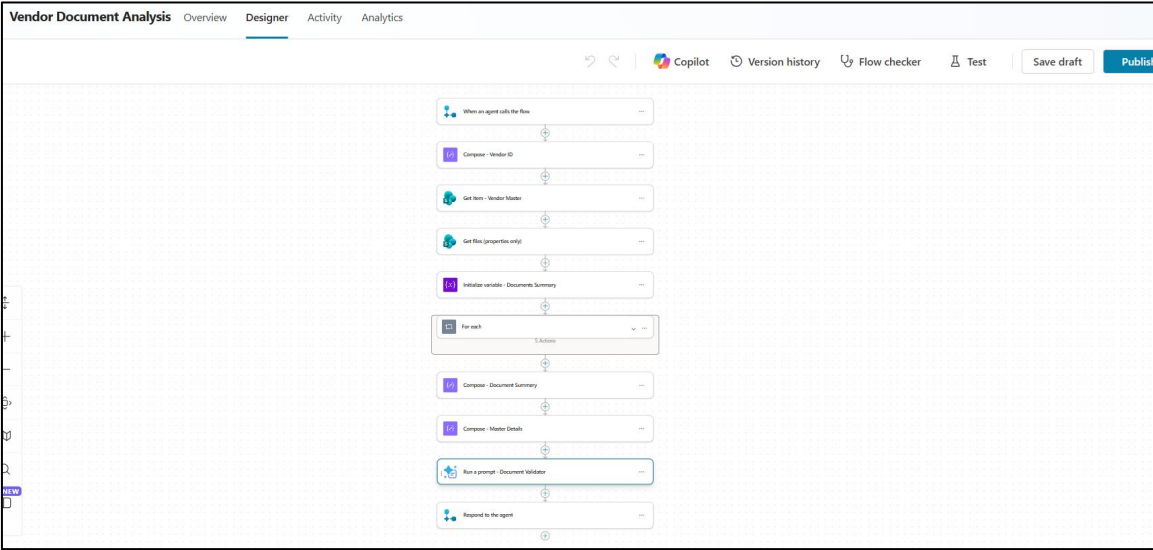
Connection *

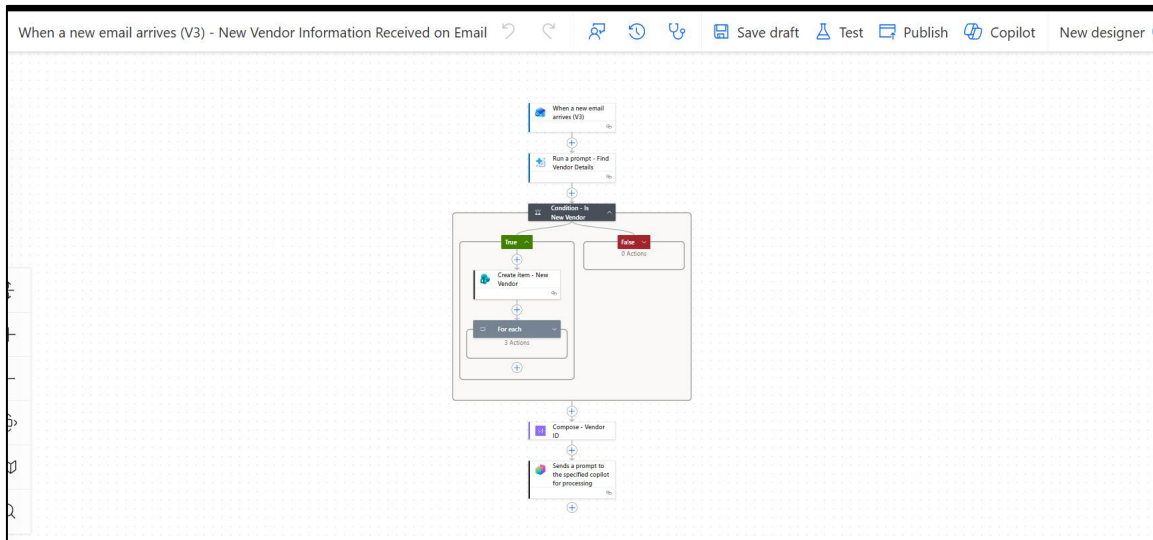
Hosted browser

Credentials to use

Maker-provided credentials

12. Agent Flows Used





13. AI Risk Scoring Engine

The AI Risk Scoring Agent aggregates findings from all specialized agents.

Risk Factors Considered

- Duplicate vendors
- Blacklist presence
- GST verification mismatch
- Invalid onboarding documents
- Missing information
- Suspicious website behavior
- Compliance inconsistencies

Output Categories

Risk Score Classification

0 – 30 Low Risk

31 – 70 Medium Risk

71 – 100 High Risk

Explainable AI

The platform generates reasoning behind the assigned score.

Example:

“Vendor classified as High Risk due to GST mismatch, missing website credibility, and duplicate vendor indicators.”

12. Technology Stack

Technology	Purpose
Microsoft Copilot Studio	AI orchestration
Power Automate	Workflow automation
SharePoint	Document storage
AI Builder	Document intelligence
Computer Use	Browser automation
Outlook	Trigger system
GPT Models	AI reasoning
Dataverse	Structured storage

13. Data Flow Architecture

Input Sources

- Vendor emails
- Attachments
- Copilot chat
- SharePoint documents

Processing Layer

- AI agents
- AI Builder
- Computer Use
- Risk engine

Storage Layer

- SharePoint
- Dataverse

Output Layer

- Email notifications
 - Investigation reports
 - Risk scores
 - Compliance recommendations
-

14. Security & Governance

The platform is designed with enterprise governance principles.

Governance Features

- Human approval workflow
- Centralized audit trail
- Role-based access
- Secure document storage
- Traceable investigation history

Responsible AI

- Human-in-the-loop review
 - Explainable AI decisions
 - Transparent risk scoring
 - Controlled automation
-

15. Human-in-the-Loop Design

The system supports responsible AI adoption.

Although investigations are automated, final approval remains with compliance and procurement teams.

Human Review Responsibilities

- Validate AI findings
- Review investigation evidence
- Approve or reject vendors
- Escalate high-risk cases

This ensures:

- Governance
 - Accountability
 - Enterprise trust
-

16. Business Impact

Operational Benefits

Metric	Improvement
--------	-------------

Metric	Improvement
Vendor Investigation Time	Reduced significantly
Manual Effort	Reduced by up to 80%
Compliance Visibility	Improved
Fraud Detection	Enhanced
Audit Readiness	Centralized tracking

Business Value

- Faster onboarding
- Reduced compliance risk
- Improved operational efficiency
- Scalable vendor verification
- Better enterprise governance

17. Demo Walkthrough

Demo Scenario

A vendor submits an onboarding request through email with attached onboarding documents.

Workflow Demonstration

Step 1

The workflow triggers automatically.

Step 2

Vendor documents are uploaded to SharePoint.

Step 3

Blacklist Detection Agent validates vendor uniqueness.

**Backlists And Duplicate Detector Age...**13.95s
Agent

Open conversation

**Document Intelligence Agent**28.45s
Agent

Open conversation

**GST Verifier Agent**116.29s
Agent

Open conversation

**Web Presence Agent**39.93s
Agent

Open conversation

**Backlists And Duplicate De...**Completed×

Agent

Description

The Blacklist & Duplicate Detector Agent validates vendor information against internal master records to identify duplicate vendors and blacklist risks. The agent checks whether a vendor with the same company name or GST Number already exists in the master database and verifies if the vendor is marked as blacklisted, restricted, or high risk. It helps prevent duplicate vendor registrations, fraud attempts, and onboarding of restricted vendors by generating automated validation and risk alerts.

Inputs

Task (String)

Validate vendor Infosys Limited with GST Number 29AAACI4798L1ZU against internal master records to identify duplicate vendors and blacklist risks.

Edit agent

⌵

Step 4

Document Intelligence Agent analyzes onboarding documents.

**Backlists And Duplicate Detector Age...**13.95s
Agent

Open conversation

**Document Intelligence Agent**28.45s
Agent

Open conversation

**GST Verifier Agent**116.29s
Agent

Open conversation

**Web Presence Agent**39.93s
Agent

Open conversation

**Document Intelligence Ag...**Completed×

Agent

Description

The Document Intelligence Agent analyzes and validates vendor-submitted documents using AI-powered document processing and metadata extraction. The agent verifies whether the uploaded documents belong to the specified vendor and extracts key business details such as GST Number, PAN Number, Bank Account Number, company name, and other relevant metadata. It helps automate document verification, reduce manual validation efforts, and identify missing, mismatched, or suspicious information during the vendor onboarding and risk assessment process.

[See less](#)

Inputs

Task (String)

Analyze and validate documents for Vendor ID 5 (Infosys Limited) to extract GST Number, PAN Number, Bank Account Number, company name, and other relevant metadata.

Edit agent

⌵

Step 5

GST Verification Agent uses Computer Use to verify GST details.

Backlists And Duplicate Detector Age... 13.95s

Agent

Open conversation

Document Intelligence Agent 28.45s

Agent

Open conversation

GST Verifier Agent 116.29s

Agent

Open conversation

Web Presence Agent 39.93s

Agent

Open conversation

GST Verifier Agent Completed

Agent

Description

This agent verifies whether a vendor's GST number is officially registered and valid on the Government GST Portal. It securely fetches and validates GST details directly from the official GST website, including business name, registration status, GST type, address, and other relevant information.

Inputs

GSTNumber (String)
29AAACI4798L1ZU

Task (String)
Verify GST registration and details for Infosys Limited with GS1

Outputs

GSTAnalysis (Record)
1 {

Edit agent

AI Vendor Risk Investigation Agent 10/05/2026, 20:18:11

AI Vendor Risk Investigation Agent 10/05/2026, 20:18:15

Good! The GST verification website has loaded. I can see a search field where I need to enter the GST Number. Let me click on the search field and enter the GSTIN: 29AAACI4798L1ZU

AI Vendor Risk Investigation Agent 10/05/2026, 20:18:18

GST Verifier Agent Completed

Agent

Description

This agent verifies whether a vendor's GST number is officially registered and valid on the Government GST Portal. It securely fetches and validates GST details directly from the official GST website, including business name, registration status, GST type, address, and other relevant information.

Inputs

GSTNumber (String)
29AAACI4798L1ZU

Task (String)
Verify GST registration and details for Infosys Limited with GS1

Outputs

GSTAnalysis (Record)
1 {

Edit agent

Step 6

Web Presence Agent validates website credibility.

Open conversation

 **Web Presence Agent**
Agent

39.93s

Open conversation

 **AI Risk Scoring Agent**
Agent

3.69s

Open conversation

 **Communication Agents**
Agent

7.91s

Open conversation

The vendor Infosys Limited with GST Number 29AAACI4798L1ZU has been validated against the internal master records. There is only one matching record, and the vendor is not blacklisted or marked as high risk. The GST certificate document is valid and matches the master record metadata. No duplicate entries or blacklist risks were found, so onboarding or further due diligence

 **Web Presence Agent** Completed
Agent

Description

The Web Presence Agent autonomously analyzes a vendor's digital footprint across public web sources to evaluate legitimacy, reputation, and operational credibility. It scans official websites, business directories, social platforms, news mentions, and online references to identify red flags such as inactive domains, suspicious content, inconsistent company information, or negative public sentiment. The agent generates structured insights that help enterprises assess vendor authenticity, trustworthiness, and overall web presence as part of the vendor risk investigation workflow.

See less

Inputs

Task (String)

Analyze web presence and digital footprint for Infosys Limited using website <https://www.infosys.com>.

Edit agent

 **AI Vendor Risk Investigation Agent** 10/05/2026, 20:20:14



 **AI Vendor Risk Investigation Agent** 10/05/2026, 20:20:20

Successfully launched the web browser and navigated to the official Infosys website at <https://www.infosys.com>



Step 7

AI Risk Scoring Agent generates vendor risk assessment.

The screenshot displays a dashboard with three agent cards: Web Presence Agent (39.93s), AI Risk Scoring Agent (3.69s), and Communication Agents (7.91s). The AI Risk Scoring Agent card is highlighted. Below the cards, a text box states: "The vendor Infosys Limited with GST Number 29AAACI4798L1ZU has been validated against the internal master records. There is only one matching record, and the vendor is not blacklisted or marked as high risk. The GST certificate document is valid and".

The AI Risk Scoring Agent interface shows a completed task: "Generate AI-based risk score for Infosys Limited (Vendor ID 5) using all available metadata and validation results." The outputs section provides a summary: "valid with metadata matching the master record. No duplicate entries or blacklist risks detected." It also lists details: "Document Intelligence: GST certificate is valid, vendor name and GST number match metadata, no duplicate or blacklist risks, PAN and Bank Account Number not found in the document." and "GST Verification: GST registration is confirmed for Infosys Limited as a Public Limited Company in Karnataka, registered since 01/07/2017, regular taxpayer, jurisdiction RANGE-".

Step 8

Communicator Agent sends final investigation report.

The screenshot displays the same dashboard as Step 7, but the Communication Agents card is highlighted. Below the cards, the same text box is present: "The vendor Infosys Limited with GST Number 29AAACI4798L1ZU has been validated against the internal master records. There is only one matching record, and the vendor is not blacklisted or marked as high risk. The GST certificate document is valid and".

The Communication Agents interface shows a completed task: "Notify stakeholders about the onboarding status and risk assessment results for Infosys Limited (Vendor ID 5)." The outputs section provides a summary: "Infosys Limited (Vendor ID 5) underwent comprehensive onboarding validation. Only one record exists for Infosys Limited with GST Number 29AAACI4798L1ZU; no duplicate or blacklist risks were detected. The GST certificate is valid and matches master metadata. GST registration is confirmed as a Public Limited Company in Karnataka, registered since 01/07/2017, regular taxpayer, jurisdiction RANGE-8SD8. The official website (https://www.infosys.com) is active,".

18. Future Enhancements

Planned Improvements

- Integration with ERP systems
 - Advanced fraud detection models
 - Real-time monitoring dashboards
 - Multi-language vendor support
-

19. Conclusion

AI Vendor Risk Investigation Agent demonstrates how autonomous multi-agent AI systems can transform enterprise vendor onboarding and compliance operations.

By combining:

- Multi-agent collaboration
- AI-powered reasoning
- Computer Use automation
- Explainable AI
- Enterprise workflow orchestration

the platform enables organizations to modernize vendor verification while improving governance, reducing risk, and increasing operational efficiency.

The solution showcases the future of enterprise AI-driven operational intelligence.